


Exam 1 on Friday

- around 12-14 questions
- a couple multiple choice, and the rest is short answer
- you do not need to write any R code
- bring a calculator!

Formulas to know:

- SSE
- RMSE
- studentized residual
- t -statistic
- confidence interval (I will provide t^*)
- equation of a fitted line, etc.

Polynomial regression

(1st-order polynomial) $y = \beta_0 + \beta_1 x + \varepsilon$

(linear)

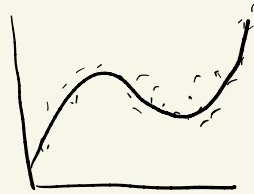


(2nd-order polynomial) $y = \beta_0 + \beta_1 x + \beta_2 x^2 + \varepsilon$

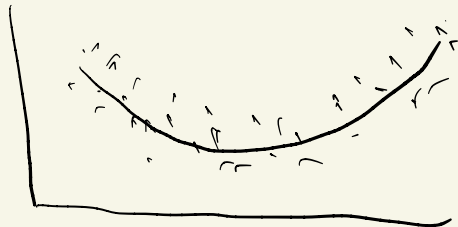
(quadratic)



(3rd-order) $y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \varepsilon$



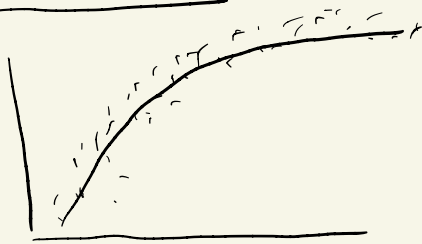
General rule: try to keep model simple, if we can



$\Rightarrow$ 2nd-order polynomial

(3rd-order unnecessary)

Other transformations



Consider log transformation:

$$y = \beta_0 + \beta_1 \log(x) + \varepsilon$$

or square root

$$y = \beta_0 + \beta_1 \sqrt{x} + \varepsilon$$

Rules for logs:

- In this course, log is always natural log (base e)
- $\log(a \cdot b) = \log(a) + \log(b)$
- $\log(a/b) = \log(a) - \log(b)$
- $\log(a^b) = b \log(a)$
- $\log(e^a) = a$

SSE { RMSE { Correlation

data $(x_1, y_1), \dots, (x_n, y_n)$, fit a model to predict y
 $\Rightarrow \hat{y}$ (fitted values)

Sum of squared errors

aka
residual
sum of squares

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

(calculate residuals, square the residuals, sum all up)

(want to minimize

SSE when fitting
model) $\rightarrow \hat{\beta}_0, \hat{\beta}_1$

The value of SSE depends on:

- the strength of relationship between x & y
- the number of observations (n)
- scale of response y

RMSE
(root mean squared error)

$$= \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

← how far our predictions
end to be from
observed response

RMSE depends on:

- strength of the relationship
- scale of response y
- not # of observations

Correlation:

$$r \in [-1, 1]$$

values close to -1 or $1 \Rightarrow$ stronger relationship

- does not depend on scale of x or y
- does depend on strength of linear relationship between x & y
(how close are the points to a best-fit line?)



← line does not capture relationship
 $r \approx 0$

Correlation is not a good choice for nonlinear relationships

SSE vs. RMSE:

- RMSE is nicer for interpretation
- we will see later that SSE is useful for hypothesis testing

Hypothesis testing: kinds of questions could include

• write down H_0 & H_A (in terms of a model parameter, e.g. β_0 or β_1)

• Calculate test statistic (given R output)

e.g.
$$t = \frac{\hat{\beta}_1 - (\text{value under } H_0)}{SE\hat{\beta}_1}$$

e.g.: $H_0: \beta_1 = 0$
$$t = \frac{\hat{\beta}_1 - 0}{SE\hat{\beta}_1}$$

t value measures how far $\hat{\beta}_1$ is from value hypothesized in H_0 , measured in standard deviations

• tell me the degrees of freedom & name of distribution

Under H_0 , $t \sim t_{n-2}$

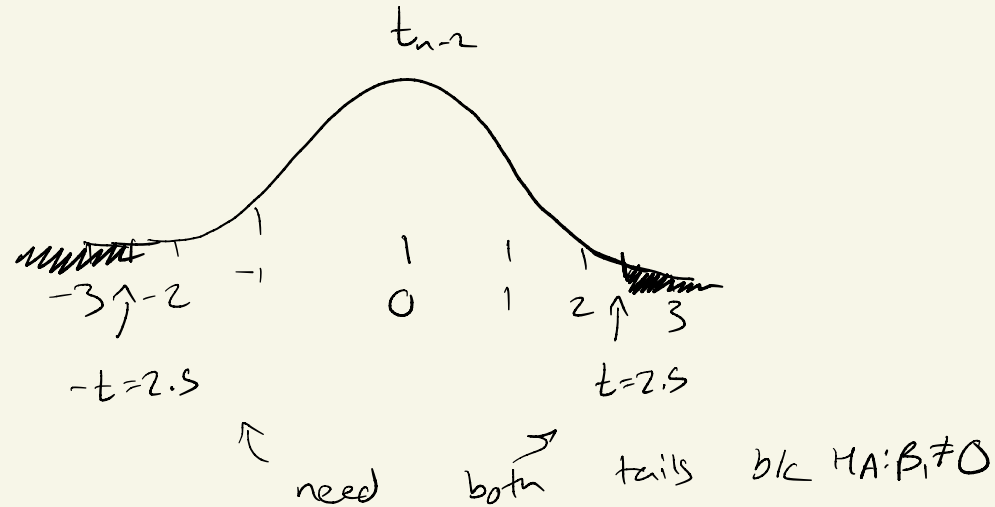
n = sample size

(for simple linear model
 $y = \beta_0 + \beta_1 x + \varepsilon$)

Shade area of t distribution corresponding to p -value

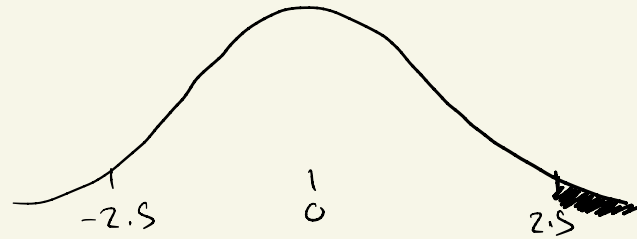
e.g. $H_0: \beta_1 = 0$ $H_A: \beta_1 \neq 0$

observed test stat : $t = 2.5$



e.g. $H_A: \beta_1 > 0$

$t = 2.5$



- I could give you p-value, and ask you to make a conclusion about hypotheses in context

Last day of material for exam: Feb 16

(more diagnostics & assumptions:

randomness,
independence,

normality $\rightarrow$ QQ plot